

# L<sup>A</sup>T<sub>E</sub>X

## Introduction Course

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# What will do today?

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- ▶ No installation
- ▶ Web browser + Overleaf account

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Git repository:

[github.com/MatsuneMikuroi/LaTeXCrashCourse](https://github.com/MatsuneMikuroi/LaTeXCrashCourse)

- ▶ This presentation (beamer)
- ▶ Script (article, multiple files)
- ▶ Compiled PDFs

# Why LaTeX?

- ▶ Open-Source and free
- ▶ High-quality typesetting
- ▶ Customizability
- ▶ Cross-platform
- ▶ Community and support
- ▶ Integration w other tools
- ▶ Overleaf

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Handle:

- ▶ Report
- ▶ Presentation
- ▶ Many more

# Code your document

Macros for:

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- ▶ Text
- ▶ Structure
- ▶ Maths
- ▶ Images and tables
  - ▶ Size
  - ▶ Placement
  - ▶ Caption
  - ▶ Label and reference
- ▶ Figures and graphs
- ▶ Bibliography
- ▶ Own commands and environments

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Full control  $\Rightarrow$  document as you want

Online editor:

# Overleaf

Online editor:

- + No installation
- + Collaboration
- + Great UI
- + Lot of docs and templates
  - Limited free features
  - Internet connection
  - Dependency on a third party

# Create your account

[overleaf.com/register](https://overleaf.com/register):

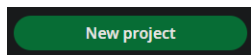
- ▶ Personal email (or Google account)
- ▶ Email confirmation

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Dashboard:



- ▶ Blank project: from scratch
- ▶ Example project: main features
- ▶ Upload project: .zip
- ▶ Templates: thesis, report, CV, poster...

# The editor

3 panels:

- ▶ File tree: new file, folder, upload
- ▶ Code editor: Code / Visual
- ▶ PDF preview

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Useful:

- ▶ Recompile: Ctrl+S / Ctrl+Enter
- ▶ Logs: errors, warnings
- ▶ Double-click PDF → code
- ▶ Share, History
- ▶ Menu: compiler, main document, download

# Writing

2 types of "text" boxes:

- ▶ Text box: no delimiter
- ▶ Math box:
  - ▶ Inline:  $\backslash(\dots\backslash)$  or  $\$...\$$
  - ▶ Display:  $\backslash[\dots\backslash]$  or  $\\$\\$\\$\\$$

# Writing

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  - ▶ Display:  $\text{\textbackslash}[\dots\text{\textbackslash}]$  or  $\text{\textbackslash}\$...\$$

2 types of macros:

- ▶ Command:  $\text{\textbackslash}\text{LaTeX}$
- ▶ Environment:  $\text{\textbackslash}\text{begin}\{\text{itemize}\}...\text{\textbackslash}\text{end}\{\text{itemize}\}$

# Preamble

Before `\begin{document}`:

- ▶ Class and options: `\documentclass{...}`
- ▶ Packages and options: `\usepackage{...}`
- ▶ Own commands and environments
- ▶ Metadata: `\title{...}`, `\author{...}`, `\date{...}`

# Minimal document

```
\documentclass{article} % preamble
\usepackage{graphicx}
\title{My title}
\author{My name}

\begin{document}          % body
\maketitle
\section{Introduction}
Hello world!
\end{document}
```

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\begin{document} % body
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Hello world!
\end{document}
```

After `\end{document}`: ignored

# Special characters

| Character | Role               | To print it     |
|-----------|--------------------|-----------------|
| %         | comment            | \%              |
| \$        | math mode          | \\$             |
| &         | table column       | \&              |
| #         | command argument   | \#              |
| -         | subscript (math)   | \_              |
| { }       | group              | \{ \}           |
| ~         | non-breaking space | \textasciitilde |
| \         | start of a command | \textbackslash  |

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Spaces:

- ▶ Many spaces = 1 space
- ▶ Line break = 1 space
- ▶ Empty line = new paragraph

# When it does not compile

Logs → **first** error first:

- ▶ Undefined control sequence: typo, missing package
- ▶ Missing \$ inserted: math outside math mode
- ▶ File ended while scanning use of...: missing }
- ▶ Environment ... undefined: missing package
- ▶ File ... not found: file name, case sensitive

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PDF despite errors ⇒ maybe wrong

# Document management

## Sectioning:

- ▶ `\part{...}`
- ▶ `\chapter{...}` (book, report)
- ▶ `\section{...}`
- ▶ `\subsection{...}`
- ▶ `\subsubsection{...}`
- ▶ `\paragraph{...}`
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## Tables of contents:

- ▶ `\tableofcontents`, `\listoffigures`, `\listoftables`
- ▶ Depth: `\setcounter{tocdepth}{...}`

# Document management (cont.)

Reference:

- ▶ `\label{...}`: sections, figures, tables, equations...
- ▶ `\ref{...}`: number
- ▶ `\pageref{...}`: page

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Multiple files:

- ▶ `\include{...}`: new page
- ▶ `\input{...}`: no new page

# Text box

Formatting:

- ▶ **Bold:** `\textbf{...}`
- ▶ *Italic:* `\textit{...}`
- ▶ Underline: `\underline{...}`
- ▶ **Colored:** `\color{red}...`

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- ▶ **Colored:** `\color{red}...`

## Environments:

- ▶ Itemize: unordered list
- ▶ Enumerate: ordered list
- ▶ Description: description list
- ▶ And many more...

# Math box

Symbols:

- ▶ Greek letters: `\alpha`, `\beta`, ...
- ▶ Operators: `\sum`, `\int`, `\frac{\dots}{\dots}`, ...
- ▶ Relations: `\leq`, `\geq`, ...
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`*`: no numbering (`equation*`, `align*`)

# Maths in practice

| Code                                                        | Result                     |
|-------------------------------------------------------------|----------------------------|
| <code>x^2</code> <code>x_i</code> <code>x_{max}</code>      | $x^2$ $x_i$ $x_{max}$      |
| <code>\frac{a}{b}</code> <code>\sqrt{x}</code>              | $\frac{a}{b}$ $\sqrt{x}$   |
| <code>\sum_{i=1}^N T_i</code>                               | $\sum_{i=1}^N T_i$         |
| <code>\bar{T}</code> <code>\Delta L</code> <code>\pi</code> | $\bar{T}$ $\Delta L$ $\pi$ |

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Braces:  $x^{10} \rightarrow x^{10}$ ,  $x^{\{10\}} \rightarrow x^{10}$

Reference: `\label{eq:...}` + `\eqref{eq:...}` (amsmath)

# Figures and tables

Figure environment:

- ▶ `\includegraphics{...}`: image
- ▶ `\caption{...}`: caption
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Table environment:

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- ▶ `&`: columns, `\\`: rows
- ▶ `\hline`: horizontal line
- ▶ `\caption{...}`, `\label{...}`

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Array: math tabular

# Bibliography

The bibliography environment:

- ▶ `\bibitem{...}`: reference
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Advanced: BibTeX, BibLaTeX

# Create figure and graphes

TikZ package:

- ▶ `tikzpicture` environment: drawing area
- ▶ Shapes, lines, text
- ▶ Flowcharts, diagrams, plots...

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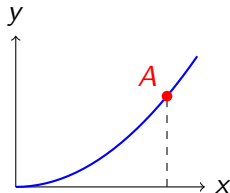
Huge package → doc, online search

# TikZ example

```
\begin{tikzpicture}
  \draw[->] (0,0) -- (2.5,0) node[right] {$x$};
  \draw[->] (0,0) -- (0,2) node[above] {$y$};
  \draw[blue, thick, domain=0:2.4]
    plot (\x, {0.3*\x*\x});
  \draw[dashed] (2,0) -- (2,1.2);
  \fill[red] (2,1.2) circle (2pt)
    node[above left] {$A$};
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Arguments:

- ▶ `[n]`: number of arguments
- ▶ `[default]`: optional first argument
- ▶ `#1, #2...`: in the definition

## Issue with macros definition

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xparse package:

- ▶ `\NewDocumentCommand{name}{arguments}{definition}`
- ▶ `\NewDocumentEnvironment{name}{arguments}{begin code}{end code}`

# Going further

## Resources:

- ▶ [Overleaf Learn](#): docs, tutorials
- ▶ [TeX Stack Exchange](#): Q&A
- ▶ [Detexify](#): drawn symbol  $\rightarrow$  command
- ▶ [CTAN](#): packages, documentation

# Conclusion

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- ▶ Academia and industry
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Overleaf:

- ▶ Online editor
- ▶ Collaboration, ease of use
- ▶ Limited free features, internet dependency